

docker_class_4(47)

creating a Dockerfile

1. sudo vim Dockerfile

FROM ubuntu

LABEL name harish

RUN touch mustafa.txt

save and continue

To build an image: docker build -t image1 .

```
[root@ip-172-31-36-247 ~]# docker build -t image1 .
Sending build context to Docker daemon 9.728kB
Step 1/3 : FROM ubuntu
latest: Pulling from library/ubuntu
aece8493d397: Pull complete
Digest: sha256:2b7412e6465c3c7fc5bb21d3e6f1917c1673584
Status: Downloaded newer image for ubuntu:latest
--> e4c58958181a
Step 2/3 : LABEL name mustafa
--> Running in a239af1e8bf2
Removing intermediate container a239af1e8bf2
--> 2ad79dc8aa3f
Step 3/3 : RUN touch mustafa.txt
--> Running in ba1497f88190
Removing intermediate container ba1497f88190
--> 94f0bc66ff70
Successfully built 94f0bc66ff70
Successfully tagged image1:latest
```

docker images

```
[root@ip-172-31-36-247 ~]# docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
image1        latest   94f0bc66ff70   13 seconds ago  77.8MB
ubuntu        latest   e4c58958181a   4 weeks ago    77.8MB
```

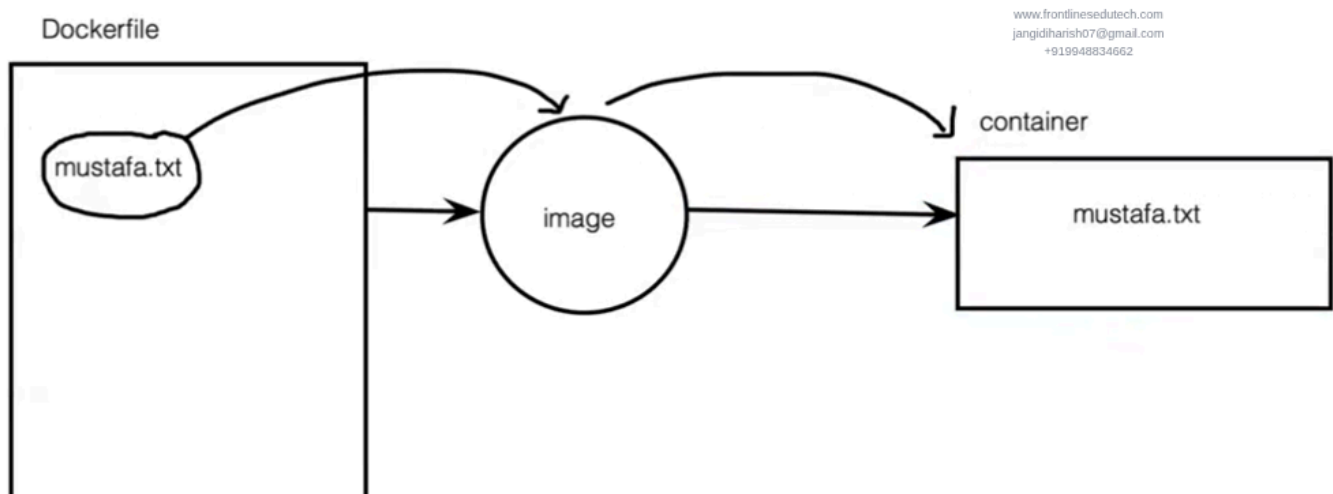
through the image i need to create container i.e

docker run -it --name cont-1 image1

enter

we are now inside the container and we can check the file caused mustafa.txt

do ls



2. sudo vim Dockerfile

FROM centos:centos7

LABEL name harish

RUN mkdir DEVOPSFROMFLM

COPY aws.txt aws.txt

To create docker image: docekr build -t image1 .

image1 already it is there so next time build it will override

locally it should created aws.txt

docekr images

```
[root@ip-172-31-36-247 ~]# docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
image1	latest	842c177e158e	7 seconds ago	204MB
<none>	<none>	94f0bc66ff70	6 minutes ago	77.8MB
ubuntu	latest	e4c58958181a	4 weeks ago	77.8MB
centos	centos7	eeb6ee3f44bd	2 years ago	204MB

image1 is latest

Create a container2

docker run -it --name cont2 image1

now we are in cont2 you can check folder and text files this is how COPY works

3. sudo vim Dockerfile

FROM centos:centos7

LABEL name FLM

RUN mkdir DEVOPSFROMFLM

ADD devops.pdf devops.pdf

and first need to create source file

build: docker build -t image2 .

it will become now again image1 latest if we dont want we give image2

create container: docker run -it --name cont3 image2

ll present now in container you can check devops.pdf and folder

what is the diff b/t copy and add

4. sudo vim Dockerfile

FROM centos:centos7

LABEL name FLM

RUN mkdir DEVOPSFROMFLM

ADD <https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.115/bin/apache-tomcat-9.0.115.zip> . // this . will be container

build: docker build -t image2:one

i want previos image2 with tag syntax is : docker build -t image2:1 . and this is like version

if you want to check is tomcat is download we have to enter container

create container: docker run -it --name cont4 image:one

```
image2 —> image2:latest
docker build -t image2:1 .
image2 —> imagename
1 —> image_tag
```

5. Sudo vim Dockerfile // WORKDIR

```
FROM centos:centos7
LABEL name FLM
RUN mkdir Harish
WORKDIR /mustafa
RUN touch harish.txt
```

build: docker build run -t image3:one .

create: docker run -it --name cont5 image3:one

```
[root@ip-172-31-36-247 ~]# docker run -it --name cont6 image3:one
[root@4f96537d6415 mustafa]#
```

6. sudo vim Dockerfile // CMD

```
FROM centos:centos7
RUN yum install git -y
build: docker build -t iamge:1
while build only it is installed git
same way if we give CMD it is also use same command
```

FROM centos:centos7

CMD yum install maven -y

build: docker build -t image:2 .

```
[root@ip-172-31-36-247 ~]# docker build -t image1:2 .
Sending build context to Docker daemon 13.82kB
Step 1/2 : FROM centos:centos7
----> eeb6ee3f44bd
Step 2/2 : CMD yum install maven -y
----> Running in 0eac5a5f3448
Removing intermediate container 0eac5a5f3448
----> 257fbccd190b
Successfully built 257fbccd190b
Successfully tagged image1:2
```

while build here it wont install maven

create: docker run -it --name cont6 image1:2

here now this time it will install amven

RUN: while build image it will install

CMD: while run the image it will perform

Note : in Dockerfile multiple RUN commands will work but multiple CMD it will not work only it will take latest1

Note: other than RUN and CMD we have entry point

7. sudo vim Dockerfile // ENTRYPOINT

FROM centos:centos7

ENTRYPOINT ["yum", "install", "git", "-y"]

build: docker build -t iamge7 .

now it will not excute when you run image it will run

create: docekr run -it --name cont7 image7

now it will start downlaod all packages i.e git

QA: MULTIPLE ENTRYPOINT WILL IT WORK ? NO it will take latest ENTRYPONT
IT WILL INSTALL ONLY LATEST PACKAGE

QA: what is the difference between ENTRYPOINT AND CMD

FROM centos:centos7

CMD ["yum", "install", "git", "-y"]

build: docker build -t iamge9 .

when image run it will install CMD I told git i ask more

create: docker run imge9 tree httpds

it will failed

if you give

docker build image9 .

it will install only one git

while cmd using extravalues not perform

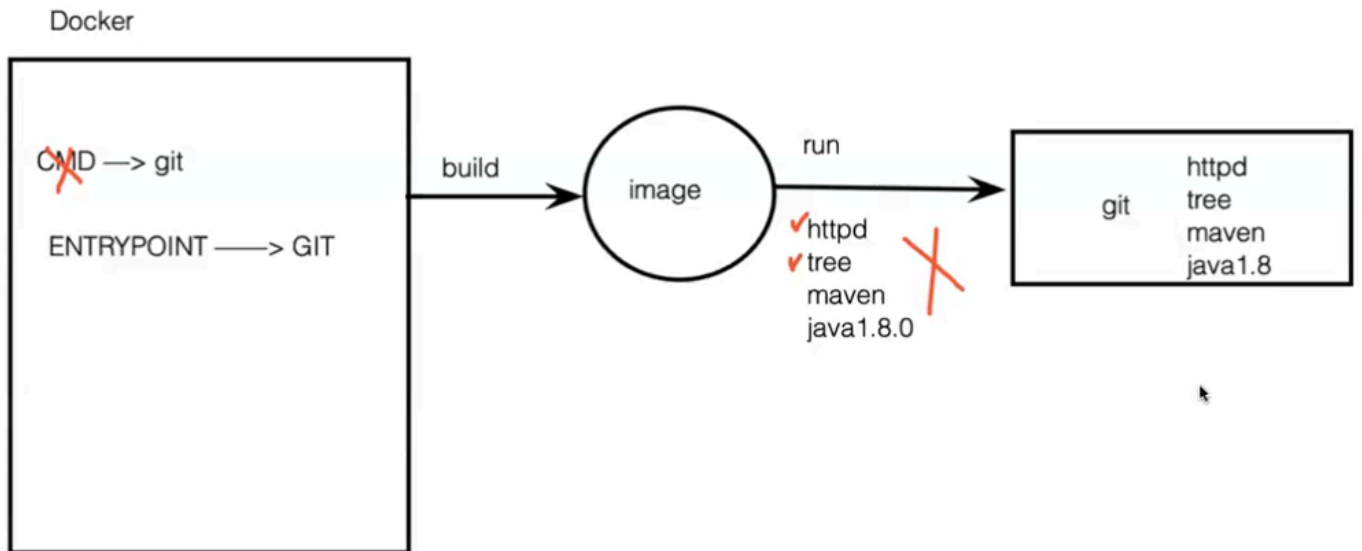
FROM centos:centos7

ENTRYPOINT ["yum", "install", "git", "-y"]

docker build -t image9 .

docker run image9 tree httpd

now it will install tree and httpd git



```

FROM centos:centos7
ENTRYPOINT ["yum", "install", "-y"]
CMD "git"
  
```

Red arrows point from the `ENTRYPOINT` and `CMD` lines to a red `XXX` mark, indicating that the `ENTRYPOINT` has higher priority than the `CMD`.

here entrypoint will have high priority

if we give entrypoint anything it will install maven if we dont mention anything and give default it will install git default

now i will pass tree i.e docker run image tree

//not this time it will install only tree

7. sudo vim Dockerfile //ENV AND ARG

```
FROM ubuntu
```

```
ARG cloud=aws
```

```
RUN echo i am learning $cloud
```

```
ENV course=devops
```

```
RUN echo i am learning $course
```

```
docker build -t iamge6 .
```

```

Removing intermediate container aa0a2be16d1
---> fabe8579e80a
Step 3/5 : RUN echo i am learning $cloud
---> Running in bfe224cde828
i am learning aws
Removing intermediate container bfe224cde828
---> 15f2614d6c39
Step 4/5 : ENV course=devops
---> Running in a16f6634023c
Removing intermediate container a16f6634023c
---> ca294e3a6721
Step 5/5 : RUN echo i am learning $course
---> Running in 9acc4944d994
i am learning devops
Removing intermediate container 9acc4944d994
---> 49f58d7550a3
Successfully built 49f58d7550a3
Successfully tagged image22:latest
[root@ip-172-31-36-247 ~]#

```

diff b/w is env and arg

arg we can override where as env we cannot override

docker build -t image44 --build-args cloud=azure

```

[root@ip-172-31-36-247 ~]# docker build -t image44 . --build-arg cloud=azure
Sending build context to Docker daemon 17.92kB
Step 1/5 : FROM ubuntu
---> e4c58958181a
Step 2/5 : ARG cloud=aws
---> Using cache
---> fabe8579e80a
Step 3/5 : RUN echo i am learning $cloud
---> Running in 272eel666d3f
i am learning azure
Removing intermediate container 272eel666d3f
---> 9a7156c43192
Step 4/5 : ENV course=devops
---> Running in a35a95cd76c0
Removing intermediate container a35a95cd76c0
---> d190f6a16c0d
Step 5/5 : RUN echo i am learning $course
---> Running in 8beff7d10b9e
i am learning devops
Removing intermediate container 8beff7d10b9e
---> 7b9e216f951d
Successfully built 7b9e216f951d
Successfully tagged image44:latest

```